Human-AI Complementarity

The cluster: four projects, one vocabulary, and the word that means three things

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Four projects sit in this cluster and they share one sentence. A human and an agent defend a network together, and I want to know whether the pair reached something neither would have reached alone. Every word in that sentence needs a definition before the sentence can be tested.

This file carries the terms that more than one of the four projects uses, and the three places where one word is doing two jobs across my own files. The rest stays with the project it belongs to: `hac_chart`, `hac_team_defense_game`, `hac_friendorfoe`, `hac_cyber_agentflow`.

Contents

1 The question underneath

What is “Human-Centered Cybersecurity”? The question is on my Headspace page with one link under it, and the link is a standards concept paper rather than a research program. That is already half the answer. The term is being fixed by an agency before the field has settled what it names.

The definition the concept paper gives:

> *“An approach that focuses on improving cybersecurity outcomes by putting people and their needs, abilities, and limitations at the forefront when designing, implementing, and making decisions about cybersecurity.”*
>
> NIST, Human-Centered Cybersecurity Guidelines and Resources Concept Paper, 2026, comment period open to 30 September 2026

The people in that sentence are the ones security is done to. The person choosing a password, the employee reporting a phish, the household configuring a device. The people in my cluster are the ones security is done by. The analyst on shift, and now the agent sitting next to her. Both start from the same refusal, that a person is a vulnerability to be contained, and both put the person back in as a defender, a reporter, a troubleshooter. They part company on which person.

The concept paper also asks a question about itself that I recognize. It asks whether human-centered cybersecurity is an approach, measured by indicators of adoption and maturity, or an outcome, requiring measures of its own such as culture, trust, usability or resilience. That is my first challenge in a different vocabulary. If it is an outcome, somebody has to measure it, and nobody has said what the measurement is. Complementarity is the version of that measurement this cluster is trying to build.

The second Headspace question is the operational one.

> *“How much “human” do we want “in the loop” of cybersecurity decision-making?”*
>
> Headspace, with a single link under it

As posed it asks for a quantity, and there is no quantity. More human is not safer and less human is not faster, because the loop is not one loop. There is the loop where an action fires, the loop where an observation is seen, and the loop where a rationale is read, and a team can be tight in one and absent in another. The answerable version is structural. Which actions require whose authorization, who sees what before it happens, and in what order. My own Challenge II already says that, and a typed dependency graph is what says it in a form a machine can enforce and an experimenter can vary.

That is the whole cluster in one line. Turn a quantity question into a structure question, then make the structure the manipulation.

2 Two challenges

The overview page states them, and everything in this lexicon descends from one or the other.

2.1 Knowledge neither side can fully state

> *"The reason a human-AI pair can beat either alone is that each holds knowledge it cannot completely externalize: the analyst's trained sense that something is off, the model's latent state."*
>
> website, Human-AI Complementarity, Challenge I

Two problems follow, and the second is the one this lexicon exists for.

> *"you cannot easily tell whether complementarity actually happened or whether one party simply deferred to the other"*
>
> website, Human-AI Complementarity, Challenge I

> *"The fundamental challenge is to make teaming a structure you can specify"*
>
> website, Human-AI Complementarity, Challenge I

A term that cannot be stated cannot be handed over, and a team whose gain cannot be attributed cannot be studied. Both halves of Challenge I are measurement problems wearing an epistemology costume.

2.2 Authority as a safety requirement

> *"An agent can act faster than a human can review; some defensive actions are irreversible; the human is accountable for what the agent does"*
>
> website, Human-AI Complementarity, Challenge II

> *"when such a team fails, the trace has to be detailed enough to say the failure was in the structure"*
>
> website, Human-AI Complementarity, Challenge II

Challenge I is why *complementarity* has three definitions across my files. Challenge II is why *trace* and *authority* have to be defined at all, and why a win-loss column is not a result.

3 The four projects

Team Defense Game

One human, one agent, one network, one attacker. Two of the five dependency types are written into the game itself, approval and a shared budget, and the agent's memory re-scores at the end of each episode against how the episode actually went. It contributes the first two dependency terms and the phrase *authority boundary*. Published in *Computers in Human Behavior: Artificial Humans* 4, 100148, in *Journal of Cybersecurity* 9(1), and at the AAAI Symposium Series 3(1).

CHART

The Configurable Human-AI Research Testbed. Team structure becomes a directed graph, the five dependency types become typed edges, and four interaction modalities sit on top of the graph. It contributes the whole dependency vocabulary, the two-layer split, and the linked record. Book chapter with CRC Press, Taylor and Francis, in *Advancements in Human Agent Teaming Research Infrastructure: Testbeds, Metrics, and Concepts*. Full entry in `hac_chart`.

FriendOrFoe

The experiment CHART was built for and the first to run on it, asking whether role dependencies raise complementarity in a team facing zero-day attacks. The design is fixed and the platform exists. It contributes the field's own definition of complementarity, the over-reliance warning, and *the seam*. Supported by the UTEP University Research Institute award. Full entry in `hac_friendorfoe`.

CyberAgentFlow

Instrumentation for a single agent driving real tools, archiving prompts, model responses, tool invocations, command outputs, analyst annotations and timing as one execution trace, then reading the trace for waste. It contributes the second sense of *trace*, and it is the project that shows why a success rate hides everything worth knowing. Extended abstract, AAAI Symposium Series 9(1), 337 to 340. Full entry in `hac_cyber_agentflow`.

There is a fifth name on the site, SOC Game, and it is a page with a title and nothing under it. What sits behind that title is a question the cluster has already put to itself twice, once on the overview page and once on the CHART artifact page: whether the graph can express how a security operations centre or an incident response team actually works. Until that is answered the SOC Game is a test of CHART's expressiveness rather than a separate project, and it is recorded here as such.

4 The shared terms

Fourteen terms, each used by at least two of the four projects. Four fields apiece. Where two of my files use one term two ways, the disagreement sits inside the entry rather than in an errata, because that is the entry worth having.

4.1 Complementarity

complementarity

> *"so that "complementarity" becomes a measurable variable"*
>
> website, Human-AI Complementarity, Challenge I

Definition: The property that a human-agent pair produced a decision neither member would have produced alone. Not joint performance, and not the pair beating one member.

Distinguishes it from: Joint performance, which is what a scoreboard reports. A pair can post the better number because one member carried it. That number is consistent with complementarity and also with total deference, and it does not separate them.

Operational test: Can you name the decision, point at the member who would not have made it alone, and say what the other member supplied? Three answers, or the claim is about the score.

In my project: The overview page treats it as a variable to be constructed. The Next page asks for it counterfactually. FriendOrFoe treats it as the dependent measure of the first experiment.

> **One word, three positions, three files**
>
> The overview page treats complementarity as something not yet measurable and sets making it measurable as the challenge.
>
> FriendOrFoe imports the field's loss-based definition and then argues that definition cannot distinguish a failed structure from an absent one.
>
> The Next page rejects both and asks for a counterfactual measure: evidence that the pair produced a decision neither the human nor the agent would have reached alone.
>
> These are not three glosses of one idea. The first says the term has no measure, the second says it has one and the measure is blunt, the third says the measure has to be counterfactual. The experiment can only pre-register one of them.

complementary team performance

"*Complementary team performance obtains when the team's loss falls strictly below both the human's and the AI's alone*"

website, FriendOrFoe, on why an aggregate score will not answer the question

Definition: The field's working definition, as stated. Complementarity holds when the empirical loss of the team is strictly lower than the minimum of the two solo losses. From Bansal, Wu, Zhou, Fok, Nushi, Kamar, Ribeiro and Weld, *Does the Whole Exceed its Parts? The Effect of AI Explanations on Complementary Team Performance*, CHI 2021.

Distinguishes it from: Complementarity as I use it above. This one is defined on aggregate loss and is therefore checkable from a results table. Mine is defined per decision and is not.

Operational test: Compute three losses on the same instances. Is the team loss strictly below both? A tie is a failure.

In my project: FriendOrFoe states it, then states the reason it is rarely reached: the region where exactly one member is correct tends to be small. A team can miss the benchmark because the structure was wrong or because no complementarity was available to find, and the aggregate does not separate the two.

4.2 Structure

interdependence

"*Interdependence is usually described and then assumed.*"

website, Team Defense Game, why build it this way

Definition: The patterned ways team members rely on one another for information, authorization, timing and execution. The cluster takes the term from organizational and team science, where the lineage runs through Thompson, *Organizations in Action*, 1967, and Cooke, Gorman, Myers and Duran, interactive team cognition, *Cognitive Science* 37, 2013.

Distinguishes it from: Cooperation, which is a disposition of the members. Interdependence is a property of the task and the authority structure, and it holds whether or not anyone wants to cooperate.

Operational test: Can one member's action change what another member is able to do, without either of them communicating? If yes, they are interdependent.

In my project: Written into the Team Defense Game as approval and a shared budget, then generalized in CHART into five typed edges that an experimenter can add and remove. That generalization is the reason CHART exists.

the structure is the manipulation

"The team structure is the manipulation, not the backdrop."

website, Human-AI Complementarity, on Team Defense Game

Definition: The cluster's design commitment. The organization of the team is the independent variable, not a fixed setting inside which other variables are studied.

Distinguishes it from: Studying trust, workload or explanation quality under one fixed team. Those results are real, and none of them tells you whether the finding survives a change in how the team is organized. That is the question CHART was built to make askable.

Operational test: Between two conditions, can you name the single edge or policy that changed? If the manipulation cannot be written down as a change to the structure, the structure is still the backdrop.

In my project: The Team Defense Game varies the authority boundary and the size of the pool. CHART turns the whole graph into the setting. Both artifact pages state the same discipline: vary one thing at a time and read the trace.

authority boundary

"This is where authority actually lives. Not in a policy document, but in whether the action fires."

website, Team Defense Game, on approval

Definition: The line between actions an agent may execute on its own and actions that require another teammate's authorization, as enforced by the software rather than as described in a protocol.

Distinguishes it from: Autonomy level, which is a property of the agent. An authority boundary is a property of the pair, and it can be tightened without touching the agent at all.

Operational test: Take the action out of the approval set and rerun. If the action fires where it previously waited, the boundary was real and you have just moved it.

In my project: In the Team Defense Game, actions carrying a control dependency sit in the pending strip until the human signs off, and expire if left alone. In CHART the boundary is an edge, and the artifact page names tightening it as the first of five single named changes an experimenter can make.

4.3 The three dependencies that cross projects

control dependency

"an action requires authorization from another agent before it executes"

website, CHART, five dependency types

Definition: An edge from a requester to an approver, carrying the set of actions that require authorization. Approval chains and escalation protocols: shutting down a production server, deploying a honeypot.

Distinguishes it from: A temporal dependency, which also blocks an action until something else happens. The blocker there is a prior action, and it can be satisfied by the requester acting alone. A control dependency can only be satisfied by another agent's decision.

Operational test: Who can clear the block? Another agent means control. A prior step in your own queue means temporal.

In my project: Named on the Team Defense Game page as the thing that makes the pending strip, listed first among CHART's five, and configured on the FriendOrFoe design under a different name. See the box below.

> **Five types on three pages, three types on the fourth, and one name that has no edge**
> The CHART research page, the CHART design page and the book chapter all name the same five: control, pool, synchrony, temporal, informational.
> The FriendOrFoe page names three, and one of them is not on that list: "which of the leader, pool and informational dependencies carry the effect".
> The platform has no edge type called leader. What the platform does have at the authorizing end of a control edge is a role, and the chapter's own input table names it `approver`. So the two pages are either describing one manipulation under two names or two different manipulations, and nothing written down decides it.
> What decides it is one line of the experiment's configuration. If the FriendOrFoe condition adds an edge whose `dependency_type` is `control`, then leader is a role name and the page should say control. If it adds anything else, there is a sixth type and CHART's five are not the set.

pool dependency

"contributions must reach a threshold before a composite action fires"

website, CHART, five dependency types

Definition: An edge from each contributor into a pool node, which holds the contributed actions until a threshold is met and then issues one composite action once. Isolating an enterprise-wide subnet takes concurrence from two of three analysts.

Distinguishes it from: A shared budget. Both make two agents affect each other without communicating, and they run in opposite directions: a pool needs contributions to accumulate before anything happens, a budget needs them to stay low so that something still can.

Operational test: Does the second contribution make the action possible or make it impossible? Possible is a pool. Impossible is a budget.

In my project: The Team Defense Game page calls the shared budget a pool dependency, which is the budget sense: "Every action either of them takes leaves less for the other". CHART's pool edge is the threshold sense, with a pool node that fires a composite action. The word is doing both jobs, and the two are mechanically opposite. The Team Defense Game entry keeps the budget sense and says so.

informational dependency

"what one agent may see of another's observations"

website, CHART, five dependency types

Definition: An edge carrying a filter over one agent's observation before another agent receives it. A junior analyst gets filtered alert logs; a senior responder integrates across sources.

Distinguishes it from: Communication. A message is something a teammate chooses to send. An informational edge is what the teammate is permitted to see whether or not anyone sends anything.

Operational test: If both agents fall silent, does one of them still see less than the other? That difference is the informational edge.

In my project: One of CHART's five, and one of the three FriendOrFoe varies. It is also the only dependency type that changes what the agent's own policy is conditioned on, which makes it the one that changes the learning problem and not only the interaction.

4.4 Evidence

trace

"the traces you need before you can attribute a failure to anything at all"

website, Human-AI Complementarity, on CyberAgentFlow

Definition: A time-stamped record of an engagement whose entries are linked to one another, so that a sequence can be reconstructed rather than inferred from separate logs.

Distinguishes it from: A log. A log answers what happened. A trace answers what happened because of what, and the difference is the linking, not the volume.

Operational test: Take one decision in the record and ask what preceded it in a different stream. If you have to reconstruct that by hand from timestamps, you have logs.

In my project: Challenge II sets the requirement: detailed enough to say the failure was in the structure. CHART meets it with cross-stream identifiers. CyberAgentFlow meets it for a single agent driving real tools.

> **Three things called a trace**
> CHART: the linked Input-Process-Outcome record of a human-AI team session, whose unit is a coordination sequence across several agents.
> CyberAgentFlow: the execution trace of one agent's tool use, whose unit is a prompt, a tool invocation and its output.

> The overview page uses the bare word for both, in one sentence about CyberAgentFlow that also states the requirement CHART is built to meet.
> Both are traces by the definition above. They differ in what the linked entities are, and therefore in what a failure can be attributed to: a structural fault in one, a wasted tool call in the other. A file that needs to be precise says team trace or execution trace.

over-reliance

"displaying the AI's explanation, confidence and label induced over-reliance"

website, FriendOrFoe, on what the seam carries

Definition: Accepting the agent's recommendation at a rate higher than its accuracy supports, including on the instances where it is wrong.

Distinguishes it from: Trust. Trust is the disposition. Over-reliance is the behaviour, and the two come apart: a person can report low trust and still approve everything, because approving is fast and checking is not.

Operational test: Split acceptance by whether the recommendation was correct. If acceptance is flat across the split, reliance is not tracking accuracy.

In my project: The finding FriendOrFoe is built to test, in the direction that matters for design: showing the evidence alone produced better calibrated reliance than showing the evidence plus the label plus the confidence. More transparency was worse. That is the reason the explanation panel is a variable in CHART and not a feature.

trust calibration

Definition: Reliance that tracks the agent's actual reliability, rather than being uniformly high or uniformly low. The lineage the chapter cites is Lee and See, *Trust in Automation: Designing for Appropriate Reliance*, Human Factors 46, 2004, and Parasuraman and Riley on misuse and disuse, Human Factors 39, 1997.

Distinguishes it from: Trust level. A high level is not a calibrated one. The calibrated quantity is the correlation between reliance and reliability, which needs at least two reliability conditions to estimate at all.

Operational test: Do you have instances where the agent was wrong, and did the human behave differently on them? Without both, trust was measured but calibration was not.

In my project: Named as the read-out in CHART's approval and explanation modalities, and it is the mechanism behind the FriendOrFoe over-reliance result. The overview page states the practical version: trust is the real bottleneck, because it decides whether the recommendation on the screen is worth acting on.

4.5 The seam

the seam

"What is handed across the seam changes the team, in a direction that is not obvious in advance."

website, FriendOrFoe

Definition: The interface between the human and the agent, considered as a design variable: what content crosses it, in what form, at what moment.

Distinguishes it from: The interface as a user-experience surface. The seam is defined by what information is transferred, not by how it is rendered, and the over-reliance result is a result about content rather than about layout.

Operational test: Name the items that cross. Evidence, label, confidence, rationale, uncertainty. Remove one and rerun. If nothing changes, that item was decoration.

In my project: FriendOrFoe treats the seam as the manipulation and CHART supplies the four modalities that constitute it. The Next page names the unfinished half: an agent surfacing its latent reasons in a form a human can correct.

tacit knowledge

"This is the tacit-to-explicit ladder, run across a human-machine seam rather than inside one head."

website, Next, on moving tacit knowledge across the boundary

Definition: The knowledge each member holds and cannot completely externalize. The analyst's trained sense that something is off, on one side; the model's latent state, on the other.

Distinguishes it from: Private information, which is knowledge one party has and could state if asked. Tacit knowledge cannot be stated on demand, which is why a protocol for moving it cannot start by demanding it be articulated.

Operational test: Ask the holder to write it down. If what comes back is thinner than what they act on, the remainder is the tacit part.

In my project: The premise of Challenge I and the reason complementarity is hard to attribute. The Next page splits it into two directions, agent to human and human to agent, and calls both open.

human in the loop

Definition: Any arrangement in which a person's decision sits between an agent's proposal and its execution. The cluster's position is that the phrase names a family of arrangements and not a dial.

Distinguishes it from: Human on the loop, where the person monitors and can interrupt but is not in the execution path. The distinction is whether the action waits.

Operational test: If the person does nothing, does the action happen? Yes is on the loop. No is in the loop. A timeout policy that auto-approves converts the second into the first without anyone editing the graph.

In my project: The Headspace question asks how much. The cluster's answer is that the quantity is the wrong shape, and the structural question replaces it. The timeout policy is where the two collide, because it is one field on one edge and it decides which of the two arrangements the team is actually running.

4.6 Load

human load

Definition: The cost the coordination structure imposes on the person, read from the record rather than only from a questionnaire: queue depth, decision latency, how often an explanation was opened, how often a proposal was modified before approval.

Distinguishes it from: Task difficulty, which belongs to the scenario. Two conditions can share a scenario and differ in load purely because one of them routes more actions through an approval queue.

Operational test: Hold the scenario fixed and change one edge. Any difference that remains is load imposed by the structure.

In my project: One of the four read-outs CHART reports alongside quality, time and error, and the reason the approval modality is described as a trade-off rather than a safeguard. Approval buys oversight and spends attention.

5 What stays in the project files

`hac_chart`: taskwork interdependence, teamwork modalities, the DAG and its node types, synchrony and temporal dependencies, all the configurable inputs, the four modalities, capability profile, the Input-Process-Outcome record, dependency check, preference data.

`hac_team_defense_game`: the pending strip, the shared budget as its own sense of pool, episode-end re-scoring, the agent memory of situation, action and blended value.

`hac_friendorfoe`: role dependency, the leader naming, zero-day as the experimental condition, the combination rule, the ten IDS scenarios.

`hac_cyber_agentflow`: execution trace, interaction lifecycle, context budget, and the five failure patterns, each of which is a term.

The check that closes this file is the two-way one. Every entry above names a nearest neighbour, and each of those neighbours has to name it back. Control names temporal and temporal names control. Complementarity names joint performance and complementary team performance names complementarity. Pool names the shared budget and the Team Defense Game entry has to name the threshold sense. Where a distinction runs one way only, one of the two definitions is wrong, and the pool pair is the one to watch.

6 Next

Three questions carry into year two, and none of them is a matter of building more platform.

> *"A team can win because it is more than its parts, or because one strong member carried it while the other watched. Win rate cannot tell these apart."*
>
> website, Next, a metric of complementarity

> *"The open question is a measure of complementarity proper: evidence that the pair produced a*

decision neither the human nor the agent would have reached alone"

website, Next, a metric of complementarity

"If the value of the pairing lies in knowledge neither side can fully state, the interface between them is the research problem."

website, Next, moving tacit knowledge across the boundary

"A fixed authority graph is a simplification."

website, Next, authority that shifts

"When that shift should happen, whether it can be learned, and how to allow it without giving up the safety the fixed structure was there to guarantee, is the third open question."

website, Next, authority that shifts

The first two are measurement. The third is a design question that the platform makes askable and does not answer: every edge in CHART today is fixed for the duration of a session, and real teams hand the wheel back and forth as the stakes move. A graph that can be rewritten mid-mission is a different safety argument from a graph that cannot, and the argument has to be made before the feature is built.

If you want the machinery rather than the vocabulary, CHART is where it lives.